

PASSWORD TOOL

Python Project Report

Submitted By:

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1. INTRODUCTION

In today's world, we use the internet for many things like email, social media, online shopping, and banking. For all these services, we need passwords to keep our accounts safe. But many people use weak passwords like "123456" or "password" which are easy to hack.

This project "Password Tool" is made to help users with password security. It can check if a password is strong or weak. It can also create new strong passwords for users.

2. OBJECTIVES

The main objectives of this project are:

- To help users check the strength of their passwords
 - To generate strong and secure random passwords
 - To spread awareness about password security
 - To learn and apply Python programming concepts
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3. ABOUT THE PROJECT

Password Tool is a command-line program made in Python language. It is simple to use and has a menu-based interface.

The project has two main features:

Feature 1: Password Strength Checker

This feature asks the user to enter a password. Then it checks the password based on five rules:

1. Length should be 8 or more characters
2. Should have lowercase letters (a-z)
3. Should have uppercase letters (A-Z)
4. Should have numbers (0-9)
5. Should have special characters (@, #, \$, %, etc.)

For each rule the password follows, it gets one point. Based on total points:

- 0-2 points = Weak Password
- 3-4 points = Medium Password
- 5 points = Strong Password

Feature 2: Password Generator

This feature creates a new random password. The user enters how long they want the password to be. The program then creates a password using random letters, numbers, and special characters.

4. MODULES USED

This project uses two Python modules:

1. random module

- This is a built-in Python module
- It is used to select random characters
- Function used: `random.choice()`

2. string module

- This is also a built-in Python module
 - It provides collections of characters
 - `string.ascii_letters` = all small and capital letters
 - `string.digits` = numbers 0 to 9
 - `string.punctuation` = special characters
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5. SOURCE CODE

```
import random
import string

# Password Strength Checker
def check_strength(password):
    score = 0
    if len(password) >= 8:
        score += 1
    if any(c.islower() for c in password):
        score += 1
    if any(c.isupper() for c in password):
        score += 1
    if any(c.isdigit() for c in password):
        score += 1
    if any(c in string.punctuation for c in password):
        score += 1
    if score <= 2:
        return "Weak Password"
    elif score == 3 or score == 4:
        return "Medium Password"
    else:
        return "Strong Password"

# Password Generator
def generate_password(length):
    characters = string.ascii_letters + string.digits + string.punctuation
    password = "".join(random.choice(characters) for _ in range(length))
    return password

# Main Program
print("PASSWORD TOOL")
print("1. Check Password Strength")
print("2. Generate Password")
choice = input("Enter your choice (1 or 2): ")

if choice == "1":
    password = input("Enter password to check strength: ")
    print("Strength:", check_strength(password))
elif choice == "2":
    length = int(input("Enter the password length: "))
    print("Generated Password:", generate_password(length))
else:
```

```
print("Invalid choice!")
```

6. OUTPUT SCREENSHOTS

Output 1: Main Menu

```
PASSWORD TOOL
1.Check Password Strenght
2.Generate Password
Enter your choice (1 or 2):
```

Output 2: Checking Weak Password

```
PASSWORD TOOL
1.Check Password Strenght
2.Generate Password
Enter your choice (1 or 2):1
Enter Password to check strenght:HII123
Strenght: weak password
```

Output 3: Checking Medium Password

```
PASSWORD TOOL
1.Check Password Strenght
2.Generate Password
Enter your choice (1 or 2):1
Enter Password to check strenght:HII12#
Strenght: medium password
```

Output 4: Checking Strong Password

```
PASSWORD TOOL
1.Check Password Strenght
2.Generate Password
Enter your choice (1 or 2):1
Enter Password to check strenght:Potato#1234
Strenght: strong password
```

Output 4: Generating Password

```
PASSWORD TOOL
1.Check Password Strenght
2.Generate Password
Enter your choice (1 or 2):2
enter the password lenght:9
generate password: 3bVRa/,g@B
```

7. ADVANTAGES

- Easy to use with simple menu options
- Helps create strong passwords
- Checks password strength instantly
- Does not store any passwords (safe to use)
- Works offline without internet

8. LIMITATIONS

- Works only in command line (no graphical interface)
- Cannot save generated passwords
- Does not check if password contains common words

9. FUTURE IMPROVEMENTS

- Add a graphical user interface (GUI)
- Add option to save passwords in a file
- Add option to copy password to clipboard
- Check password against common password lists

10. CONCLUSION

Password Tool is a simple but useful Python project. It helps users understand the importance of strong passwords. The project uses basic Python concepts like functions, loops, conditions, and modules.

This project taught me how to:

- Write functions in Python
- Use built-in modules
- Take user input and show output
- Write clean and organized code

I believe this tool can help many people create better and stronger passwords for their online accounts.

11. BIBLIOGRAPHY

- Python Documentation: <https://docs.python.org>
 - W3Schools Python Tutorial: <https://www.w3schools.com/python>
 - GitHub: <https://github.com>
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Thank You